

Sr. No.	Topic Name
1	Scalar and vector fields, Differentiation of Vector functions
2	Formula for finding gradient, Geometrical meaning as normal vector, Formulas to find divergence and curl of vector fields, Illustrative examples
3	Defining Line integrals as the limit of a finite sum and work done by a force field, Methods to evaluate Line integrals, Illustrative examples
4	Statement of Green's Theorem, Application to evaluate Line integrals, Evaluating plane area, Illustrative examples
5	Statement of Stokes's theorem, Based numericals
6	Statement of Gauss's theorem, Application to find surface integral over Cubical and spherical surfaces only, Jacobian with simple problems
7	Definition of vector space, Linear independence and dependence of vectors,
8	Basis and dimension
9	Linear transformations, Range and Kernel
10	Rank and nullity of Linear transformation
11	rank- nullity theorem(only statement) and its applications
12	Composition of Linear map, Inverse of Linear Transformation
13	Matrix associated with linear map
14	Revision of definition of field, and define inner product space (for complex and real field both) with examples
15	Define orthogonal sets and orthonormal sets with examples.
16	Define Gram-Schmidt orthogonalizations process and Solve problems related to its application.
17	Defining first order exact differential equation, necessary and sufficient condition, General solution.
18	n th order homogeneous linear equation $f(D)y=0$, linear independence of solutions, auxiliary equation, solution: when roots of auxiliary equation are a) distinct, b) equal, c) complex.
19	n th order non-homogeneous linear equation $f(D)y=r(x)$, general solution = Complimentary function+ particular integral, method to find PI when $r(x)=e^{ax}$.
20	Method to find PI when $r(x)=\sin ax, \cos ax$. And $r(x)=x^n$,
21	method to find PI when $r(x)=e^{ax}V(x)$, $x\sin ax, x\cos ax$
22	Variation of parameter method to find PI of a second order linear differential equation.
23	Cauchy-Euler equation and its solution.
24	Finding solution of a system of linear equations.
25	Mathematical modeling. Basic elements of an electric circuit, Kirchhoff's law, solution of simple LR and CR circuits
26	Basic concept and classification of second order PDE
27	Separation of variable method to solve second orders linear homogeneous PDEs with constant coefficients.
28	One dimensional wave equation as mathematical model of vibrations of a stretched string, solution of 1-dim wave equation by SOV method, Solution of 1-dim wave equation with different initial conditions.
29	One dimensional heat equation as mathematical model for the temperature distribution in a thin heated rod. Solution of 1-dim heat equation with both ends of the rod at 00 . Solution of 1-dim heat equation when the rod has insulated ends.
30	Two dimensional Laplace equation as a mathematical model for the steady state temperature distribution in a thin rectangular plate, solution of the equation, Solution of Laplace equation with different boundary conditions.